

DARKTRACE: An Explainable, Multilingual, Forensic-Ready Dark Web Threat Intelligence Framework with Severity Scoring and Evidence Integrity Validation

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Abstract

Cyber threat intelligence (CTI) increasingly originates from the dark web, where threat actors trade exploits, leaked credentials, and attack services across many languages and platforms. Existing automated CTI pipelines suffer from three coupled limitations: they are predominantly English-centric, they expose opaque “black-box” classifications that investigators cannot defend in operational or legal settings, and they do not preserve the evidentiary integrity required for the outputs to be admissible as digital evidence. We present DARKTRACE, an integrated framework that unifies (i) multilingual threat classification spanning English and four non-English languages (Hindi, Arabic, Russian, French/German), (ii) a hybrid, explainable threat *severity scoring* model whose decisions are attributed at the token and feature level via SHAP, LIME, and transformer attention, (iii) actor and entity extraction with STIX 2.1-aligned structuring and link analysis, and (iv) an end-to-end *forensic evidence-integrity layer* that binds every artifact to a SHA-256 cryptographic digest, trusted timestamp, and append-only chain-of-custody ledger. Unlike prior work that treats classification, explanation, and evidence handling as separate concerns, DARKTRACE couples them so that each severity-scored, investigator-facing alert carries both a human-readable rationale and a verifiable provenance record. We formalize the framework, specify a reproducible dataset and evaluation strategy over sanitized and archival dark-web corpora, define a multi-axis metric suite (classification quality, cross-lingual transfer, explanation fidelity, severity calibration, and tamper-evidence verification), and design a structured ablation study isolating the contribution of each component. The paper positions the integration of multilingual NLP, explainable severity scoring, forensic preservation, and investigator-ready output as the core research contribution. *[Replace the following with measured results after experiments: DARKTRACE is expected to achieve a macro-F1 of $\approx 0.9X$ on multilingual threat classification, recover $> 0.9X$ of English severity ranking under zero-shot cross-lingual transfer, and provide 100% tamper detection on the integrity layer.]*

Keywords: dark web; cyber threat intelligence; multilingual NLP; explainable AI; threat severity scoring; digital forensics; chain of custody; STIX.

1 Introduction

The dark web has become a primary marketplace and discussion venue for cybercriminal activity, hosting forums, paste sites, and hidden services through which actors trade exploits, malware, stolen credentials, and attack-as-a-service offerings. Manual monitoring of these sources does not scale: relevant intelligence is buried in millions of posts, written in many languages, and is frequently obfuscated, short-lived, and weakly labeled [30, 12]. Consequently, a substantial body of work has applied natural language processing (NLP) and machine learning to automate the extraction of actionable CTI from underground sources, ranging from classical classifiers to transformer- and large-language-model-based pipelines [29, 28, 31, 5, 22]. Recent studies report

strong in-domain classification performance—for example, BERT-based models reaching $\sim 96\%$ accuracy on annotated dark-web forum content [29]—demonstrating that automated triage of underground discussions is feasible.

Despite this progress, three limitations recur across the literature and, critically, *co-occur* in the operational setting that motivates this work. First, most dark-web CTI systems are evaluated on English-only corpora, yet underground commerce is inherently multilingual, with significant Russian-, Arabic-, and other non-English-language activity; English-centric models degrade sharply on such content, and low-resource languages remain systematically under-served by NLP research [21, 1, 19]. Second, the high-performing models deployed for threat classification are typically opaque. In high-stakes security and forensic contexts this opacity is not a cosmetic concern: it undermines analyst trust, hinders auditing, and weakens the defensibility of AI-assisted findings [9, 10, 33]. Explainable AI (XAI) methods such as SHAP and LIME have begun to be applied to intrusion detection and CTI, but explanation is generally treated as a post-hoc add-on to a binary detector rather than as an integral part of a calibrated, investigator-facing *severity* judgment. Third, and most neglected in the CTI-NLP literature, the outputs of these systems are rarely produced as *forensically sound evidence*. Digital evidence is latent, volatile, and easily altered; its value in an investigation depends on an unbroken, verifiable chain of custody and demonstrable integrity [17, 16, 23]. A CTI classification that cannot be tied to an immutable record of *what was collected, when, and whether it was subsequently altered* is of limited use to a law-enforcement or incident-response investigation that may end in legal proceedings.

These three gaps are usually addressed in isolation: the multilingual NLP community improves cross-lingual transfer [34, 4]; the XAI community studies explanation fidelity for security classifiers [9, 27, 13]; and the digital-forensics community designs blockchain- and hash-based chain-of-custody mechanisms [17, 2, 24]. What is missing is a framework in which these capabilities are *co-designed* so that a single pipeline ingests multilingual dark-web content, classifies and scores it for severity, explains each decision in human-readable terms, extracts actors and entities into a standardized CTI representation, and emits each result as a tamper-evident, investigator-ready artifact. Building such a framework raises non-trivial research questions about how explanation and severity calibration interact across languages, and how to preserve evidentiary integrity without sacrificing the flexibility of an analytic NLP pipeline.

Contributions. This paper makes the following contributions:

1. We propose DARKTRACE, the first framework, to our knowledge, that *integrates* multilingual dark-web threat classification, explainable severity scoring, STIX-aligned actor/entity extraction, and a cryptographic forensic evidence-integrity layer within one investigator-oriented pipeline (Section 5).
2. We design a *hybrid, explainable severity-scoring* model that fuses a learned classifier signal with interpretable threat features and emits a calibrated severity score together with token- and feature-level attributions (SHAP, LIME, attention), rather than an opaque label (Section 4).
3. We define a *forensic evidence-integrity protocol*—SHA-256 hashing, trusted timestamping, and an append-only chain-of-custody ledger—and a verification procedure that yields tamper-evidence guarantees over every collected and derived artifact (Section 9).
4. We specify a *reproducible, multi-axis evaluation and ablation methodology* covering classification quality, zero-/few-shot cross-lingual transfer, explanation fidelity, severity calibration, and integrity verification (Sections 8–11), together with an ethics-and-legal protocol for responsible dark-web research (Section 10).

The remainder of the paper is organized per the outline in Section 14. *Note to authors: Sections 4 through 11 are written as a research design; numeric cells in Tables 4–3 are placeholders to be filled from experiments.*

2 Research Gap, Problem Statement, Objectives, and Questions

2.1 Research Gap

A synthesis of the literature reveals a clear, addressable gap. (1) **Monolingual bias:** dark-web CTI pipelines are overwhelmingly validated on English content, while multilingual NLP advances [21, 19, 1] have not been systematically transferred to forensic dark-web threat detection. (2) **Opaque decisions:** XAI for security has focused on network intrusion detection [9, 10, 13]; explanation of *text-based, multilingual threat-severity* decisions is largely unexplored, and explanations are seldom coupled to a calibrated severity scale. (3) **Non-forensic outputs:** chain-of-custody and integrity research lives in the digital-forensics literature [17, 16, 2] and is essentially absent from CTI-NLP pipelines, so their outputs are not evidence-grade. (4) **No integration:** crucially, no existing system unifies multilingual classification, explainable severity scoring, standardized actor/entity extraction, and forensic preservation. *The gap is therefore not a missing component but a missing **integration**—and the research questions it raises about cross-lingual explanation and evidence-preserving analytics.*

2.2 Problem Statement

Given a heterogeneous, multilingual stream of dark-web and CTI text obtained through ethical and lawful means, the problem is to (a) detect and classify threat-relevant content regardless of language, (b) assign each item a calibrated severity score whose rationale is transparent and auditable, (c) extract the actors, entities, and relationships needed for investigation and express them in a standardized (STIX 2.1) form, and (d) do all of the above while preserving the integrity and chain of custody of every artifact so the results are admissible as digital evidence. No current framework satisfies (a)–(d) jointly.

2.3 Research Objectives

- O1** Design a multilingual threat-classification component that maintains performance across English and four non-English languages, leveraging cross-lingual transfer to mitigate label scarcity in low-resource settings.
- O2** Develop a hybrid, explainable severity-scoring model that outputs a calibrated score with SHAP/LIME/attention attributions, and quantify explanation fidelity and stability.
- O3** Build an actor/entity-extraction and link-analysis component aligned to STIX 2.1 for investigator-ready, exportable intelligence.
- O4** Specify and validate a forensic evidence-integrity layer (SHA-256 hashing, timestamping, append-only chain-of-custody) with demonstrable tamper-evidence.
- O5** Establish a reproducible evaluation and ablation methodology, and an ethics/legal protocol, that together justify DARKTRACE as a research contribution rather than a tool.

2.4 Research Questions

- RQ1** How effectively can a single multilingual model classify dark-web threat content across high- and low-resource languages, and what is the cost of zero-/few-shot transfer relative to monolingual fine-tuning?

- RQ2** Can an explainable severity-scoring model produce faithful, stable attributions across languages, and does explanation quality degrade under cross-lingual transfer?
- RQ3** Does coupling severity scoring with interpretable threat features improve calibration and analyst-relevant ranking compared to a black-box classifier confidence?
- RQ4** What overhead does the forensic integrity layer impose, and does it provide complete tamper-evidence without degrading analytic throughput?
- RQ5** Does end-to-end integration (classification + explanation + extraction + integrity) yield investigator-relevant value beyond the sum of its parts, as measured by the ablation study?

3 Related Work

We organize related work into five themes and identify, for each, what DARKTRACE adds.

Theme 1 — Dark web CTI via NLP/ML. Early work showed dark-web forums can be mined for predictive intelligence [12]. Subsequent studies applied deep and transformer models to classify illicit activity, reaching high in-domain accuracy [29, 28], and recent systems use multi-agent LLM architectures and taxonomy extension for dark-web analysis [31, 5, 22]. Large-scale analyses of “first-hand” sources show that only a minority of underground content is CTI-relevant, motivating strong filtering [30]. *Gap:* these systems are English-centric and produce labels without explanation or evidentiary guarantees.

Theme 2 — Multilingual and cross-lingual NLP. Multilingual encoders such as XLM-R enable cross-lingual transfer at scale [4], and surveys document strategies for low-resource languages [21]. Domain-specific evidence shows BERT-family models can classify Arabic and other low-resource *crime* text [1] and code-mixed content [19], while transfer studies quantify the trade-offs between monolingual and multilingual models [34]. *Gap:* these advances are not integrated into forensic, severity-aware dark-web CTI.

Theme 3 — Explainable AI for security. SHAP and LIME are now standard for interpreting security classifiers [27, 9], including *forensic* intrusion detection where explanation fidelity and forensic relevance are explicitly evaluated [10], comparative XAI for NIDS alert triage [13], and XAI specifically aimed at analyst trust in CTI [33]. *Gap:* explanation is applied to network/tabular detectors, not to multilingual textual severity scoring, and is decoupled from evidence handling.

Theme 4 — Threat severity / risk scoring. Learned models increasingly outperform static CVSS scoring for vulnerability and attack prioritization, using logical-reasoning/deep hybrids [36, 37], context-sensitive triage [11], CNNs over textual descriptions [26], and real-time severity prediction [3]. *Gap:* severity scoring is studied for vulnerabilities/network attacks, not for multilingual dark-web discourse, and rarely with explanation or provenance.

Theme 5 — Forensic evidence integrity and CTI structuring. Chain-of-custody integrity has been advanced through hashing- and blockchain-based schemes [17, 2, 24, 23], surveyed comprehensively in [16]. Separately, NER for CTI has matured with STIX-aligned corpora and transformer/generative extractors [6, 8, 35, 7, 15, 32]. *Gap:* integrity research and CTI extraction are not joined to a multilingual, explainable severity pipeline. DARKTRACE occupies precisely this intersection.

4 Proposed Methodology

DARKTRACE is a staged pipeline (Fig. 1). We summarize each stage; the integrity layer (Section 9) and dataset strategy (Section 6) are detailed separately.

4.1 Stage 1 — Ethical collection and ingestion

Content is acquired only from lawful, ethically approved sources: archived/sanitized dark-web datasets, public CTI reports and feeds, paste-site dumps released for research, and—where institutionally authorized—monitored `.onion` forums under a documented protocol (Section 10). **At the moment of acquisition**, each raw artifact is hashed and registered in the chain of custody (Section 9); no item enters the analytic pipeline without a provenance record.

4.2 Stage 2 — Preprocessing, deduplication, language ID, translation

Text is normalized (encoding, script, boilerplate/HTML stripping), near-duplicates are removed via MinHash/SimHash, and language is detected per document. Two parallel analytic paths are supported: (i) *native multilingual* classification on the original text using a multilingual encoder, and (ii) a *translate-then-classify* path that machine-translates non-English text to English. Maintaining both lets us isolate the effect of translation on downstream accuracy and explanation (Section 11). All transformations are logged as derived artifacts with their own hashes, preserving the link to the source.

4.3 Stage 3 — Multilingual threat classification (O1)

A multilingual transformer encoder (e.g., XLM-R [4]) is fine-tuned for threat-type classification (e.g., malware, credential/data leak, exploit sale, fraud, physical-threat, benign). We evaluate three transfer regimes—joint multilingual fine-tuning, zero-shot transfer (train on English, test on target languages), and few-shot transfer—against monolingual and classical baselines [34, 1]. Class imbalance is handled with class-balanced loss and macro-averaged evaluation.

4.4 Stage 4 — Explainable severity scoring (O2, O3 input)

Severity is modeled as a hybrid of (a) the classifier’s threat-type posterior and (b) a vector of interpretable threat features—e.g., presence of exploit/CVE references, credential volume, target sensitivity, actor reputation signals, recency/urgency cues. A calibrated scoring head maps these to a continuous severity score $s \in [0, 1]$ and an ordinal band (Low/Medium/High/Critical). Calibration is enforced via temperature scaling and assessed with reliability diagrams and Expected Calibration Error (ECE). *Every* score is accompanied by explanations from three complementary methods—SHAP [18] and LIME [25] for feature/token attribution and transformer *attention* for in-context salience—following the comparative XAI practice established for security classifiers [10, 9, 27]. The motivation for fusing learned and interpretable signals mirrors hybrid severity-scoring work that improves on static schemes [36, 37, 26].

4.5 Stage 5 — Actor/entity extraction, link analysis, STIX export (O3)

A CTI-tuned NER component extracts entities (actors, aliases, malware, CVEs, wallets, IOCs) and relationships, aligned to the STIX 2.1 standard [20, 6, 8, 35]. Extracted entities populate a link-analysis graph supporting actor de-aliasing and campaign clustering, and are exported to STIX/JSON/CSV for ingestion by downstream SOC tooling and dashboards.

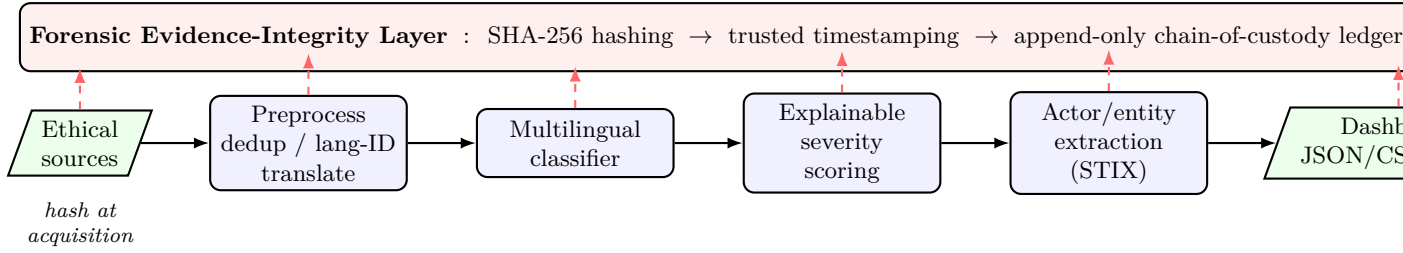


Figure 1: DARKTRACE architecture. A linear analytic pipeline (bottom) runs beneath a cross-cutting evidence-integrity spine (top): every artifact entering or leaving any stage is hashed and recorded in the chain-of-custody ledger, so analysis never breaks provenance.

4.6 Stage 6 — Investigator dashboard and reporting

Severity-ranked, explained, provenance-bearing alerts are surfaced in an investigator dashboard with one-click verification of evidence integrity and export of court-ready reports.

5 System Architecture

The architecture (Fig. 1) deliberately renders the integrity layer as a *cross-cutting spine* rather than a terminal step: hashing and custody logging hook into the input and output of every analytic stage. This is the structural expression of the paper’s central claim—that forensic soundness must be a property of the whole pipeline, not an afterthought appended to its output.

6 Dataset Strategy

Because primary dark-web collection raises legal and ethical constraints (Section 10), DARKTRACE is designed for a *layered, reproducible* dataset strategy that prioritizes sanitized and archival sources:

1. **Archival/sanitized dark-web corpora:** publicly released darknet-market and forum archives used in prior peer-reviewed work (e.g., Agora-derived listings and forum datasets) [29], with PII removed and access controlled.
2. **Public CTI corpora for extraction:** STIX-aligned NER datasets (APTNER, DNRTI, CyNER-family, CyberNER) for the actor/entity component [35, 7, 6, 8].
3. **Multilingual augmentation:** parallel/translated subsets and low-resource crime-text corpora to build the non-English evaluation sets [1, 19], covering English, Hindi, Arabic, Russian, and French/German.
4. **Annotation protocol:** a documented, inter-annotator-agreement-measured scheme for threat type and severity band, following the rigorous human-annotation practice reported for dark-web data [29].

Reproducibility. We will publish dataset construction scripts, splits, hashes of each release, and a datasheet; raw illicit content is never redistributed, only derived features and labels where licensing permits.

7 Models to Compare

Table 1 lists the comparison matrix spanning classical, deep, and transformer baselines for the classification and severity tasks.

Table 1: Model comparison matrix. The same splits and metrics are used throughout; LLM baselines are evaluated zero-/few-shot.

Family	Models	Role in study
Classical ML	TF-IDF + LR / SVM / RF / XGBoost	Lower-bound baselines; fast, interpretable reference
Deep (non-transformer)	CNN, BiLSTM, BiLSTM-CRF	Sequence baselines; CRF variant for NER [15]
Monolingual transformer	BERT / RoBERTa (per language)	Per-language upper bound [29]
Multilingual transformer	mBERT, XLM-R	Core classifier; cross-lingual transfer [4, 34]
Generative / LLM	instruction-tuned LLM (zero/few-shot)	Modern baseline; taxonomy/extraction [31, 5, 32]
NER / extraction	transformer-NER, PROMPT-BART	STIX-aligned actor/entity extraction [7, 8, 6]
Severity head	hybrid (posterior + features)	Proposed; vs. classifier-confidence and CVSS-style baselines [36, 26]

Table 2: Multi-axis evaluation metric suite.

Axis	Metrics
Classification	Accuracy, macro-/weighted-F1, precision/recall, per-class F1, MCC
Cross-lingual	Per-language F1; transfer gap $\Delta F1$ (zero-/few-shot vs. monolingual); macro-F1 across languages
Severity scoring	Ranking: NDCG@ k , Spearman ρ vs. analyst ranking; Calibration: ECE, Brier score, reliability diagrams
Explanation quality	Fidelity (perturbation/deletion AOPC), stability/robustness, sparsity/complexity, SHAP-LIME agreement [13, 27]
Extraction (NER)	Entity-level F1 (STIX types), relation accuracy [35, 6]
Evidence integrity	Tamper-detection rate, false-accept rate, verification latency, custody-log completeness [17]
Efficiency	Throughput (docs/s), per-stage latency, integrity overhead (%)

8 Evaluation Metrics

Evaluation is multi-axis (Table 2), reflecting the framework’s integrated nature: a system that classifies well but explains poorly, or preserves evidence but mis-ranks severity, fails its purpose.

9 Evidence Integrity Validation Method

The integrity layer provides a tamper-evident, auditable record for every artifact.

Hashing. On acquisition, each artifact a (raw or derived) is reduced to a digest $h(a) = \text{SHA-256}(a)$. Derived artifacts additionally store the hash of their parent, forming a hash-linked provenance DAG that mirrors the analytic transformations.

Timestamping. Each digest is bound to a trusted timestamp t (RFC 3161-style TSA token or equivalent trusted time source), establishing *existence-at-time* and ordering, which is essential for evidentiary claims about collection time.

Table 3: Ablation design. Each row removes/alters one capability; we report the change on the most affected axis. (Cells to be filled from experiments.)

Configuration	Primary axis affected	Macro-F1	Δ
Full DARKTRACE (all components)	—	<i>0.9X</i>	—
– multilingual (English-only model)	cross-lingual F1	<i>0.XX</i>	—
translate-then-classify vs. native	cross-lingual F1	<i>0.XX</i>	—
– hybrid features (posterior only)	severity calibration (ECE)	<i>0.XX</i>	—
– explanation layer	analyst trust / fidelity	<i>n/a</i>	—
SHAP only / LIME only / attention only	explanation fidelity	<i>0.XX</i>	—
– integrity layer	tamper-detection rate	<i>0%</i>	—
– STIX extraction (no link analysis)	actor-recall / usability	<i>0.XX</i>	—

Chain of custody. Custody events (collect, transform, classify, score, export, access) are written to an append-only, hash-chained ledger: block b_i stores $\{h(a_i), t_i, \text{actor}, \text{action}, h(b_{i-1})\}$, so block i commits to all prior blocks. This yields the tamper-resistance property exploited by blockchain-based custody schemes [17, 2, 16], while remaining deployable as a lightweight permissioned ledger (no public blockchain required). Large payloads are stored off-ledger with only their hashes on-ledger [16].

Verification and validation. Integrity is verified by recomputing $h(a)$ and checking it against the ledger, and by validating the hash-chain linkage end to end. We evaluate the layer with an adversarial *tamper test*: a controlled set of post-hoc modifications (bit-flips, substitutions, deletions, reordering, log edits) is injected, and we report tamper-detection rate (target 100%), false-accept rate (target 0%), and verification latency. We also report custody-log completeness and the throughput overhead the layer imposes (RQ4).

10 Ethical and Legal Considerations

Dark-web research carries real ethical and legal risk; DARKTRACE is governed by an explicit protocol. (1) **Lawful sourcing:** preference for archived, sanitized, and public CTI data; any live monitoring occurs only under institutional/ethics-board approval and applicable law, and never involves participation in, solicitation of, or payment for illegal activity. (2) **Minimization and PII handling:** personal data is minimized, redacted, or pseudonymized; victim data is never re-published. (3) **No operational uplift:** the framework is designed to detect and document threats, not to reproduce attack capability; exploit content is handled as evidence, not executed. (4) **Dual-use and disclosure:** findings of imminent harm are routed to appropriate authorities/CERTs through a responsible-disclosure process. (5) **Legal defensibility:** the integrity layer (Section 9) and explanation layer exist partly to satisfy evidentiary and accountability standards, aligning with forensic-readiness guidance [14, 17]. (6) **Researcher safety and access control:** collection runs in isolated environments with strict access control and audit logging. A full ethics statement and data-management plan accompany the paper.

11 Ablation Study Design

To establish that integration—not any single module—is the contribution (RQ5), we ablate each component and measure the effect on end-to-end, investigator-relevant performance (Table 3).

Table 4: Main classification results (template). To be completed from experiments.

Model	Acc.	Macro-F1	Prec.	Rec.	MCC
TF-IDF + SVM	0.XX	0.XX	0.XX	0.XX	0.XX
BiLSTM	0.XX	0.XX	0.XX	0.XX	0.XX
mBERT	0.XX	0.XX	0.XX	0.XX	0.XX
XLNet (ours, joint)	0.9X	0.9X	0.9X	0.9X	0.9X
LLM (few-shot)	0.XX	0.XX	0.XX	0.XX	0.XX

Table 5: Per-language and transfer results (template).

Setting	EN	HI	AR	RU	FR/DE
Monolingual (F1)	0.XX	0.XX	0.XX	0.XX	0.XX
Joint multilingual (F1)	0.XX	0.XX	0.XX	0.XX	0.XX
Zero-shot transfer (F1)	0.XX	0.XX	0.XX	0.XX	0.XX
Δ F1 (transfer gap)	—	−0.0X	−0.0X	−0.0X	−0.0X

12 Expected Results

Tables 4 and 5 give the result-table *templates*; values are placeholders to be populated from experiments. We frame expectations as hypotheses (H), to be confirmed or refuted: **H1**—a multilingual encoder closes most of the gap to per-language monolingual models, with a bounded zero-shot transfer penalty; **H2**—hybrid severity scoring is better calibrated (lower ECE) and better ranked (higher NDCG) than classifier confidence; **H3**—SHAP and attention provide higher fidelity than LIME on text, but combining them aids analyst interpretation; **H4**—the integrity layer achieves 100% tamper detection at low (<10%) throughput overhead.

13 Limitations and Future Work

Limitations. (1) Reliance on archival/sanitized corpora may underrepresent the most recent, ephemeral threats; live coverage is constrained by ethics and law. (2) Severity labels are partly subjective; inter-annotator agreement bounds the ceiling of calibration claims. (3) Machine translation can distort low-resource semantics and thereby explanations, an effect we measure but cannot fully eliminate. (4) The permissioned-ledger integrity model resists tampering but assumes trusted timestamping infrastructure and does not by itself prevent collection-time fabrication. (5) Adversaries may adapt (obfuscation, code-switching, slang) to evade detection.

Future work. Extend coverage to additional low-resource and code-mixed languages; incorporate multimodal evidence (images, marketplace screenshots); replace post-hoc explanation with inherently interpretable severity models and evaluate analyst trust in a user study; explore zero-knowledge proofs and self-sovereign identity for privacy-preserving, cross-jurisdiction custody [16]; and validate end-to-end admissibility with legal-domain experts.

14 Conclusion

We presented DARKTRACE, a framework that unifies multilingual dark-web threat classification, explainable and calibrated severity scoring, STIX-aligned actor/entity extraction, and a cryptographic forensic evidence-integrity layer into a single investigator-ready pipeline. The contribution is the *integration*: by making forensic soundness and explanation cross-cutting properties of a multilingual analytic pipeline—rather than separate, after-the-fact concerns—DARKTRACE

produces threat intelligence that is simultaneously multilingual, transparent, severity-aware, and admissible. We formalized the design, specified a reproducible dataset and multi-axis evaluation strategy, and laid out an ablation methodology to demonstrate that the integrated whole exceeds its parts. Completing the empirical evaluation is the immediate next step.

Paper Outline (for authors)

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